

Enhanced Removal of the Light Reflection of Eyeglass Using Multi-Channel CycleGAN with Difference Image Equivalency Loss

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ABSTRACT

Eye gaze estimation rapidly becomes the required technology in virtual reality environments. The geometrical model base method is commonly used by equipping infrared (IR) light sources and cameras inside of the head mount displays (HMDs). Some HMDs enable users to wear with spectacles on, and, in this case, IR light reflections of the eyeglass often causes serious obstruction to detect those of the corneal. In a previous study, we had proposed the multi-channel CycleGAN to generate the eye images with no eyeglass from those with eyeglass. It seemed to be successful but not perfect especially in serious eyeglass reflection cases. In this study, we additionally applied the difference image equivalency loss and late fusion of channels to improve removal performance of eyeglass reflections and reduce mixing actions across channels, which achieved enhanced removal of the eyeglass reflections.

Keywords: Virtual reality, gaze estimation, generative adversarial network, reflection removal.

Index Terms: Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality; Computing methodologies—Machine learning—Machine learning algorithm

1 INTRODUCTION

Virtual reality (VR) and augmented reality (AR) systems rapidly became practical and familiar in the consumer market. Recently, the high-end head mount displays (HMDs) equipped with 4K (3840x2160) or higher resolutions and 90 Hz or higher frame rates have been increasing. Consequently, many of them support the foveated rendering, which takes advantage of the capability of human visual perception. It projects higher resolution graphics only in the area that a user is staring and makes lower resolution in the peripheral area, so that it effectively achieves higher sense of reality with lighter graphical computation. Therefore, the gaze estimation plays a much more important role in recent HMDs.

Eye gaze estimation can be classified into two categories: the geometrical model base and the appearance base methods. In VR environments, the former method is commonly used by equipping infrared (IR) light sources and cameras in HMDs. Then, the gaze estimation is normally accomplished by calculating the positional relation between apparent corneal reflections of IR light sources and the specific reference points of the eye, i.e., the center of the pupil. However, some HMDs enable users to wear with spectacles on, and, in this case, the eyeglass reflections often cause serious obstruction for detecting the corneal reflections.

In the previous study [1], we had proposed the multi-channel CycleGAN. CycleGAN [2] is one of the major evolved generative adversarial networks (GANs) [3], which achieved image-to-image

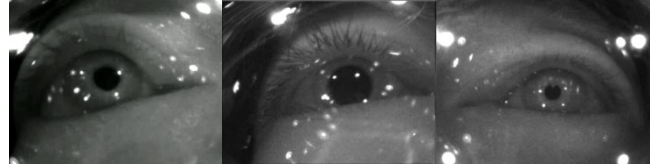


Figure 1: Example images of eyeglass reflection.

translation by learning the respective common features in two categorized image groups. It performed transferring or replacing features, e.g., generating oil paintings from still images or replacing horse to zebra. We applied this technique for generating the eye images with no eyeglass from those with eyeglass. Moreover, by using multiple eye images of different time points, the previous study achieved to selectively remove the impassive eyeglass reflections and keep the active corneal reflections alive in sequential eye images. In this study, we additionally applied the difference image equivalency loss to improve removal quality of the impassive reflections. We also applied the late fusion of channels to reduce mixing actions across them. Both contributed to achieve the enhanced removal of the eyeglass reflections.

2 METHOD

CycleGAN learns the feature relation between two image groups (X and Y), and generates generators ($G: X \rightarrow Y$ and $F: Y \rightarrow X$) and discriminators (D_X and D_Y). In this study, X is the eye images with wearing glasses and Y is those with no glasses. Figure 1 shows the example images of X , where serious obstructive reflections of the eyeglass are observed in addition to the four corneal reflections around the iris. The target of this study is to remove only the eyeglass reflections and reserve the corneal reflections.

Loss functions used in this study are expressed as (1), where L_{GAN} is the adversarial loss, L_{cyc} is the cycle consistency loss, L_{id} is the identity mapping consistency loss, and L_{eq} is the proposed difference image equivalency loss.

$$L(G, F, D_X, D_Y) = L_{GAN}(G, D_Y, X, Y) + L_{GAN}(F, D_X, Y, X) + L_{cyc}(G, F) + L_{id}(G, F) + L_{eq}(G, F) \quad (1)$$

Then, the difference image equivalency loss is expressed as:

$$L_{eq}(G, F) = \sum_{i=1}^n \sum_{j=i+1}^n (D_{i,j}(G) + D_{i,j}(F)), \quad (2)$$

where

$$D_{i,j}(G) = |(X_i - G(X_i)) - (X_j - G(X_j))|, \quad (3)$$

$$D_{i,j}(F) = |(F(Y_i) - Y_i) - (F(Y_j) - Y_j)|, \quad (4)$$

X_i and Y_i denote the channel i images, and n is the number of input channels. Since G generates the no-eyeglass images, we can ideally assume that $X_i - G(X_i)$ will become the equivalent images specifying real eyeglass reflections for all channels. Similarly, F generates the eyeglass images so that $F(Y_i) - Y_i$ will become the equivalent images specifying generated eyeglass reflections. Thus, reducing the proposed loss value L_{eq} is expected to work for improving removal performance of the eyeglass reflections.

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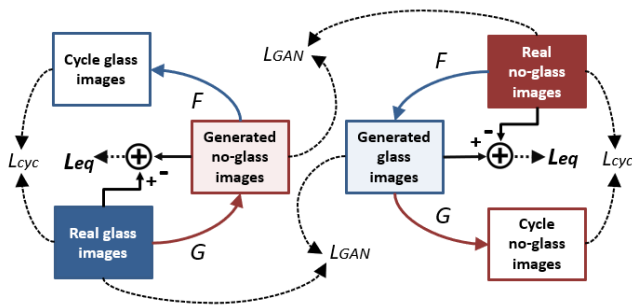


Figure 2: Configuration of CycleGAN used in this study.

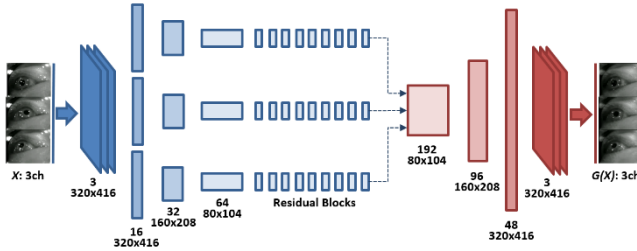


Figure 3: Generator design using late channel fusion.

Figure 2 illustrates the configuration diagram of the CycleGAN used in this study, where the difference image calculation parts are highlighted, and the identity mapping parts are eliminated for visibility. Figure 3 illustrates the generator design. In the previous study, we used four channel (ch) images which comprised three images at the different time points and one blink image. In this study, we stopped relying on the blink image and only used former three images ($n=3$). In addition, we applied the late fusion of channels so that the network is separated for each channel in the earlier encoder stage and fused in the later decoder stage. This design avoids the mixing actions across multiple channels in the earlier encoder stage and is expected to work for reducing a chance of mixing actions as a total generator system.

3 EXPERIMENT

We used the same dataset as the previous study: the images with wearing glasses (X) and those with no glasses (Y) of ten people (five males and five females). Here, X and Y has consecutive 500 frames for each people; we define this unit as one chunk so that X and Y has ten chunks, respectively. In addition, data augmentation by flipping and rotating (within $\pm 10^\circ$) images was applied for training. In this case, the same flipping and rotation angle was applied for all 3 ch input images. We utilized the training batch size of 4, which is also the same as the previous study. In training, the 3 ch input images were comprised of three frames randomly sampled out from one chunk. For testing, we separately prepared the videos, where the current frame and the randomly sampled two frames from the past were used as the 3 ch input images.

4 RESULTS

Figure 4 shows the example results associated with the images shown in Figure 1. The proposed method achieved good removal performance. Detailed comparison is shown in Figure 5, where (a) original image, (b) applied previous method, (c) applied proposed method and (d) calculated eyeglass reflections $X_i - G(X_i)$, are indicated. Visual observation reveals that the proposed method enhanced the removal performance of the eyeglass reflections. Also, the quantitative comparison between the difference image $X_i - G(X_i)$ and the artificially created eyeglass reflection image figured out the improved removal quality than the previous study.



Figure 4: Generated no-eyeglass images.

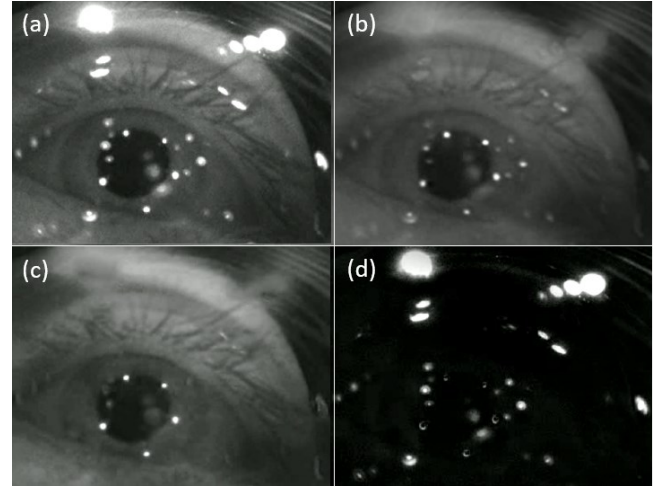


Figure 5: Comparison of removal performance. (Brightness: $\times 1.3$)

5 DISCUSSION

Since the 3 ch input images will have different corneal reflections potentially, the equivalency of $X_i - G(X_i)$ for all channels seemed to encourage them to converge to the same eyeglass reflection images; and $F(Y_i) - Y_i$ as well. Applying this difference image equivalency loss for the cycle images may improve the removal performance furthermore. Meanwhile, the late fusion of channels, i.e., channel fusion in the last three convolutional layers, seemed to correctly work for distinguishing the moving reflections of corneal and the stationary reflections of eyeglass. However, as far as we experimented, the later channel fusion, i.e., last two layers or one, could not achieve satisfactory performance. Besides, the visual evaluation using sequential eye images figured out that the late channel fusion worked effectively for reducing mixing actions.

6 CONCLUSION

The difference image equivalency loss and the late fusion of channels were applied for the multi-channel CycleGAN; and it achieved enhanced removal of the stationary eyeglass reflections. They also contributed to reduce the mixing actions across multiple channels. Besides, one channel reduction from the previous study was succeeded by discarding the blink image.

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